

Linear Search Algorithm:

It is also known as sequential search.

It is the simplest searching algorithm used to find a target value within a data collection.

arr = [18, 12, 9, 14, 77, 50]

Q) Find whether 14 exists in array or not!
If no value found, return -1

Time Complexity:

Best case

$O(1)$

Worst case

$O(N)$

$N \Rightarrow$ size of array

Best case scenario :

If the target element is at the first index of the array.

e.g. $arr = [8, 9, 12, 18, \dots, n \text{ elements}]$

target = 8

- * 1 comparison is made and the loop is executed once.
- * Doesn't matter the size of the array.

Worst case scenario :-

- * you do not find target

item.

* Iterate the every element and then in the end it says it doesn't find the element.

Size of array $\Rightarrow$

100	-	100 comparisons
200	-	200 comparisons
n	-	n comparisons

$O(N)$